



VCU

College of Engineering

MD-024

Vision-Based Settling Rate
Analysis for Water Treatment
Team Contract

Prepared for
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ChemTreat

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Step 1: Get to Know One Another. Gather Basic Information.

Task: This initial time together is important to form a strong team dynamic and get to know each other more as people outside of class time. Consider ways to develop positive working relationships with others, while remaining open and personal. Learn each other's strengths and discuss good/bad team experiences. This is also a good opportunity to start to better understand each other's communication and working styles.

<i>Team Member Name</i>	<i>Strengths each member bring to the group</i>	<i>Other Info</i>	<i>Contact Info</i>
Aubrey Carey	Lab Research Experience Coding Industry Experience (Process Controls)		careyae2@vcu.edu 703-853-2511
Rohan Sharma	Coding Great Questions! Debugging App Design		sharmar16@vcu.edu 571-831-1762
Virginia Anderson	Communication skills Chemistry Some Water treatment knowledge		andersonvc2@vcu.edu (757)-947-4942
Ronald Menendez	Coding Problem solving skills		menendezrd@vcu.edu (804)-432-6759

<i>Other Stakeholders</i>	<i>Notes</i>	<i>Contact Info</i>
Dr. McGill Dr. Thomson-McInnes	<i>Faculty Advisors</i>	mcgillc2@vcu.edu btmcinnes@vcu.edu
Ronald Byamungu Praise Adebisi	<i>Project Management Advisors</i>	ronaldb@vcu.edu adebisip@vcu.edu
Vlad Djukanovic Will Henderson	<i>Sponsors</i>	vladimird@chemtreat.com williamh@chemtreat.com

Step 2: Team Culture. Clarify the Group's Purpose and Culture Goals.

Task: Discuss how each team member wants to be treated to encourage them to make valuable contributions to the group and how each team member would like to feel recognized for their efforts. Discuss how the team will foster an environment where each team member feels they are accountable for their actions and the way they contribute to the project. These are your Culture Goals (left column). How do the students demonstrate these culture goals? These are your Actions (middle column). Finally, how do students deviate from the team's culture goals? What are ways that other team members can notice when that culture goal is no longer being honored in team dynamics? These are your Warning Signs (right column).

Resources: More information and an example Team Culture can be found in the Biodesign Student Guide "Intentional Teamwork" page ([webpage](#) | [PDF](#))

<i>Culture Goals</i>	<i>Actions</i>	<i>Warning Signs</i>
Constructive Feedback	● Set aside time in team meetings to review one another's work and offer suggestions	Editing other's work without consultation
Be proactive	● Setting personal goals of what needs to be done by next group meeting ● Doing tasks in parallel and working together to ensure deadlines are met ● Not leaving things till the last minute	Missing deadlines Holding up the team Not showing up to meetings
Clear Communication	● Communicate workload via individual and group action items ● Organizing files ● Ensuring teammates understand the work you're doing	Not responding in text groupchat/discord

Step 3: Time Commitments, Meeting Structure, and Communication

Task: Discuss the anticipated time commitments for the group project. Consider the following questions (don't answer these questions in the box below):

- What are reasonable time commitments for everyone to invest in this project?
- What other activities and commitments do group members have in their lives?
- How will we communicate with each other?
- When will we meet as a team? Where will we meet? How Often?
- Who will run the meetings? Will there be an assigned team leader or scribe? Does that position rotate or will same person take on that role for the duration of the project?

Required: How often you will meet with your faculty advisor, where you will meet, and how the meetings will be conducted. Who arranges these meetings?

See examples below.

<i>Meeting Participants</i>	<i>Frequency Dates and Times / Locations</i>	<i>Meeting Goals Responsible Party</i>
Students Only	Once a week, on Tuesdays via Online calling and video	Update group on day-to-day challenges and accomplishments Assign or delegate work needed for project to progress Actively work on project
Students + Faculty advisor	Once a week most weeks, on Fridays at 11 am in W101	Update faculty advisor and get answers to our questions Set up/update calendar with hard deadlines
Project Sponsor	TBD Meet in person or by zoom	Update project sponsor and make sure we are on the right track

Step 4: Determine Individual Roles and Responsibilities

Task: As part of the Capstone Team experience, each member will take on a leadership role, *in addition to* contributing to the overall weekly action items for the project. Some common leadership roles for Capstone projects are listed below. Other roles may be assigned with approval of your faculty advisor as deemed fit for the project. For the entirety of the project, you should communicate progress to your advisor specifically with regard to your role.

- **Before meeting with your team**, take some time to ask yourself: what is my “natural” role in this group (strengths)? How can I use this experience to help me grow and develop more?
- **As a group**, discuss the various tasks needed for the project and role preferences. Then assign roles in the table on the next page. Try to create a team dynamic that is fair and equitable, while promoting the strengths of each member.

Communication Leaders

Suggested: Assign a team member to be the primary contact for the client/sponsor. This person will schedule meetings, send updates, and ensure deliverables are met.

Suggested: Assign a team member to be the primary contact for faculty advisor. This person will schedule meetings, send updates, and ensure deliverables are met.

Common Leadership Roles for Capstone

1. **Project Manager:** Manages all tasks; develops overall schedule for project; writes agendas and runs meetings; reviews and monitors individual action items; creates an environment where team members are respected, take risks and feel safe expressing their ideas.
Required: On Edusourced, under the Team tab, make sure that this student is assigned the Project Manager role. This is required so that Capstone program staff can easily identify a single contact person, especially for items like Purchasing and Receiving project supplies.
2. **Logistics Manager:** coordinates all internal and external interactions; leads in establishing contact within and outside of organization, following up on communication of commitments, obtaining information for the team; documents meeting minutes; manages facility and resource usage.
3. **Financial Manager:** researches/benchmarks technical purchases and acquisitions; conducts pricing analysis and budget justifications on proposed purchases; carries out team purchase requests; monitors team budget.
4. **Systems Engineer:** analyzes Client initial design specification and leads establishment of product specifications; monitors, coordinates and manages integration of sub-systems in the prototype; develops and recommends system architecture and manages product interfaces.
5. **Test Engineer:** oversees experimental design, test plan, procedures and data analysis; acquires data acquisition equipment and any necessary software; establishes test protocols and schedules; oversees statistical analysis of results; leads presentation of experimental finding and resulting recommendations.
6. **Manufacturing Engineer:** coordinates all fabrication required to meet final prototype requirements; oversees that all engineering drawings meet the requirements of machine shop or

vendor; reviews designs to ensure design for manufacturing; determines realistic timing for fabrication and quality; develops schedule for all manufacturing.

<i>Team Member</i>	<i>Role(s)</i>	<i>Responsibilities</i>
Aubrey Carey	Project Manager	• Develop overall schedule for project • Runs meetings and makes overview slides • Reviews and monitors individual action items
Virginia Anderson	Lead Experimenter (ChemE); Sponsor Contact	• Lead person for design of experiments • Manages lab notebook and data collection • Establish contact with ChemTreat • Document meeting minutes
Rohan Sharma	Lead Developer (CS); Faculty Advisor Contact	• Manages the GitHub • Schedule meetings, send updates, and ensure deliverables are met for faculty advisors • Mentor team members by providing guidance and support to help team members improve their skills
Ronald Menendez	Financial Manager	• Research technical purchases and acquisitions • Conduct pricing analysis and budget justifications on proposed purchases • Carries out team purchase requests and monitors the budget

Step 5: Agree to the above team contract

Team Member: Aubrey Carey

*Signature: **Aubrey Carey***

Team Member: Virginia Anderson

*Signature: **Virginia Anderson***

Team Member: Rohan Sharma

*Signature: **Rohans***

Team Member: Ronald Menendez

*Signature: **Ronald Menendez***

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